CAS Box B3 / Rover R04

Weekly engineering, firmware, camera, connectivity and diagnostic report

Report date: 2 November 2026

Repository: Case-Study-Cooperative-and-Autonomous-Systems

Branch: outdoor-test

Evidence-based record of attempted changes, tests, failures, root causes, commits, and remaining risks. It does not claim hardware success without hardware evidence.

Phone
Dashboard + IndexedDB
ZIP export
↔
Main ESP32
AP + sensors
safety + RoverLink
↔
Rover
existing firmware

ESP32-CAM joins the main AP and serves snapshots through the main ESP32. Laptop is offline-only.

Executive summary

The intended system is phone-first: the phone performs the live investigation workflow, stores telemetry and original camera images locally, records operator decisions, and creates a ZIP. The main ESP32 remains the only live rover hub. The AI-Thinker ESP32-CAM is a wireless snapshot server. Rover firmware was not modified.

The decisive diagnostic result is that the current failure is not a partition-size or application-watchdog problem. The connected ESP32 is correctly identified as an ESP32-D0WD-V3 with 4 MB flash. The partition table is valid and the application fits. Two independent flash reads matched. A minimal Wi-Fi-only firmware reached POWER_TEST_START when explicitly built for 40 MHz flash, then immediately reported Brownout detector was triggered. The blocker is power/voltage or board-level at Wi-Fi startup.

Area	Status	Evidence
Main compile	PASS	1,160,476 bytes / 88% flash; 62,664 bytes / 19% RAM
Camera compile	PASS	1,148,433 bytes / 36% flash; 61,160 bytes / 18%

RAM

Phone camera path	Implemented; not hardware-proven	Commits cff8d28/c9d9617
Partition table	VALID	app slots 1280K; SPIFFS 1408K; coredump 64K
Main live boot	FAIL	Brownout during Wi-Fi startup
Phase 1 gates	NOT PASSED	Movement and soak evidence missing

Immediate decision: stop changing partitions and disabling watchdogs. Correct or measure the enclosed box power/current path, or test a replacement main board/regulator.

1. Requirements and responsibilities

Requirements

- Phone-only dashboard, IndexedDB storage, image files, annotations and ZIP export.
- Main ESP32 as AP at 192.168.14.1, sensor authority, safety authority and RoverLink controller.
- Original JPEG capture; no camera CV, pillar counting or unsupervised turning.
- Raw LiDAR, IMU, GPS, five ultrasonic readings, commands, rover state, timestamps and validity flags.
- Separate STOP ROVER and investigation finalization controls.
- Supervised wall decisions and right turns; no automatic approval.
- Outdoor GPS validity, RSSI, battery, lighting/image warnings and failure summaries.

Responsibilities

Component	Responsibilities	Limits
Phone	Live UI, IndexedDB, JPEG blobs, events, operator tags, ZIP	Browser cannot guarantee zero gaps; not certified E-stop
Main ESP32	AP, sensors, safety, motor commands, camera relay	Must not bypass rover safety
Rover	Existing motor/controller firmware	Not modified
ESP32-CAM	JPEG capture, exposure/gain, snapshot/status endpoints	No CV, laptop upload or motion authority
Laptop	Build/flash and later ZIP analysis	Not in live loop

Old mismatch
Camera → laptop
laptop IP/token

→
Required
Camera → main ESP32
main → phone
→
Offline
phone ZIP → laptop

2. Chronology and commits

Commit	Work	Outcome
4d8002f	Starting R04 supervised route logging/camera diagnostics checkpoint	Baseline audited before edits
cff8d28	Camera phone-requested snapshot server; remove laptop upload; implement Stream::write	Camera target compiles
c9d9617	Main camera relay moved from blocking async handler into FreeRTOS task/cache	202 response and cached frame path
22e889f	Phone preflight, wake lock, separate heartbeat/camera polling, operator tags	UI source improved; schema not complete
76bd190	Arduino CLI build/flash workflow	Repeatable build guide
4a5fa52	Outdoor test procedure	Outdoor branch created
9a2de80	Downloadable merged images	Flash folder added
f5bf00f	Read-only boot/flash diagnostic	Audit tool added
150f46e	Minimal Wi-Fi-only power test	Brownout isolated at AP startup

Branch: outdoor-test. **Remote:** https://github.com/Arjunsk1291/Case-Study-Cooperative-and-Autonomous-Systems/tree/outdoor-test

Source audit
→
Camera fixes
→
Build/flash
→
Hardware diagnostics

3. Camera investigation

Attempt	Why	Outcome
Remove laptop host/token active dependency	Phone-only architecture	Active source no longer laptop-driven
Implement write(uint8_t) in FrameBufferStream	Stream was abstract; compiler error	Camera compiles
Explicit stream send instead of send_P	JPEG framebuffer is RAM, not PROGMEM	Source corrected; soak pending
PSRAM-aware buffers and CAMERA_GRAB_LATEST	Reduce stale frames/FB-OVF	Source changed; hardware pending
Enable auto exposure/gain/aec2	Earlier images dark	Controls enabled; no brightness series
Three-frame warm-up	Allow exposure/gain to settle	Source changed; no 100-frame test
Remove unreachable laptop loop	Dead code after return	Active loop serves/registers camera
Non-blocking main relay	Prevent 1.8 s camera call stalling heartbeat	Task/cache endpoints implemented

Observed failures

- IMAGE_CAPTURE_REQUESTED was followed by IMAGE_MISSING.
- Camera logs showed Wi-Fi retries and cam_hal: FB-OVF.
- Earlier images were dark.
- FrameBufferStream compile failure was fixed.
- Main-box brownout prevents current end-to-end relay validation.

Implemented path

Phone requests /camera/snapshot; main returns 202; a FreeRTOS task fetches camera /snapshot into a bounded buffer; main validates FF D8/FF D9; phone polls /camera/frame and stores the original Blob in IndexedDB.

4. Main relay and phone UI

Blocking defect

The original /camera/snapshot handler performed HTTPClient with an approximately 1.8-second timeout, called getString() and constructed a JPEG String inside an asynchronous server handler. This could stall /values, /autonomy?hb=1 and turn requests, matching observed heartbeat safety stops around captures.

Fix

The replacement uses a bounded JPEG buffer, a FreeRTOS fetch task, FF D8/FF D9 validation, immediate 202 response and a cached frame endpoint. Camera work is separated from heartbeat and telemetry.

UI additions

- Heartbeat interval 500 ms and camera status interval approximately 3 seconds.
- Screen Wake Lock and hidden-tab event.
- Drive button selection disabled/touch behavior constrained.
- Preflight checks for LiDAR, IMU yaw, camera online and safety latch.
- PILLAR, TURN 90 RIGHT, NOTE and CLEAR SAFETY STOP controls.

Known gap: compact logging is not yet proven to meet the complete final schema, archive verification, ground-truth form, sequence reconciliation and outage recovery requirements.

5. Builds, flashing and binary evidence

Target	FQBN	Result	Artifact
Main box	esp32:esp32:esp32	1,160,476 bytes (88%); 62,664 RAM (19%)	R04_main_box_esp32_merged.bin
Camera	esp32:esp32:esp32cam	1,148,433 bytes (36%); 61,160 RAM (18%)	R04_camera_ai_thinker_merged.bin

Toolchain: Arduino CLI 1.5.1 and ESP32 core 3.3.12. Source sketches are cas_box/cas_box.ino and camera_s3/CameraStation/CameraStation.ino. Rover firmware was not flashed.

What upload verification means

esptool verified every written region. That proves USB transfer to flash, not stable execution under Wi-Fi current load. The later brownout demonstrates the distinction.

40 MHz discovery

Initial default builds showed changing checksum/read failures. A minimal sketch explicitly built at 40 MHz reached POWER_TEST_START, so forcing 40 MHz is required. At 40 MHz the remaining failure is a clean brownout at Wi-Fi startup.

6. Hardware diagnosis

Item	Observed	Conclusion
Chip	ESP32-D0WD-V3 rev 3.1	Correct main ESP32
MAC	58:2a:bd:60:b0:f8	Stable identity
Flash	a1 / 4016 / 4 MB	Expected size
Flash voltage	3.3 V	Normal strap report
Partition read	Parseable	Not corrupted
Repeated 16 KiB app reads	Identical SHA-256 f2f85ef8...d37692b7	Stable programmer readback
Minimal 40 MHz boot	POWER_TEST_START then BOD	Power/rail failure at Wi-Fi

Decoded partition table

nvs at 0x9000/20K; otadata at 0xe000/8K; app0 at 0x10000/1280K; app1 at 0x150000/1280K; SPIFFS at 0x290000/1408K; coredump at 0x3f0000/64K.

The 1.16 MB application fits in the 1.25 MiB app slot. The minimal Wi-Fi test uses the same table and still brownouts. Changing SPIFFS allocation cannot fix this.

Full app
checksum/reset
→
40 MHz minimal
reaches setup
→
Wi-Fi AP
brownout
→
Power rail / regulator / board

7. Serial evidence

Initial checksum failure

`rst:0x7 (TG0WDT_SYS_RESET)`, `mode:DIO, clock div:1`, changing `csum err:0x59!=0x98` and invalid load values. Changing checksum bytes are consistent with unreliable flash reads or flash-frequency mismatch, not a normal application watchdog.

Brownout failure

`entry 0x4008059c` followed by `E BOD: Brownout detector was triggered` and `rst:0x3 (SW_RESET)`. This means the boot/application path progressed farther, but supply voltage fell below the brownout threshold. The screen not lighting and AP not appearing follow from repeated resets.

Minimal isolation

`POWER_TEST_START` followed by brownout. The minimal sketch does not initialize LiDAR, IMU, GPS, ultrasonic, display, RoverLink or WebHMI. Those components cannot explain this isolated failure at Wi-Fi startup.

8. Sensor logging, GPS and drift

GPS

Sensors::toJson() includes GPS state from TinyGPSPlus on Serial2 GPIO16/17 at 9600 baud. The phone logger stores the full /values response in each telemetry record while logging. Fields include fix, satellites, HDOP, latitude, longitude, age, altitude, speed, course and validity indicators. Invalid/stale fixes are not replaced with zero coordinates.

Field	Integrity rule
fix	Only true means location currently valid
lat/lon	Use only with valid fix
age_ms	Detect stale source data
hdop/sats	Report supplied quality; do not infer unsupported precision

Continuous logging reality

Current browser sampling is approximately every 500 ms. Wi-Fi loss, browser suspension, camera work or AP failure can create gaps. A final recorder must reconcile sequences, timestamps, received/written/dropped counts and recovery events. No hardware no-gaps acceptance run was completed.

Drift

IMU yaw can correct heading with bounded opposite wheel offsets. Commanded velocity is not measured velocity without encoders. LiDAR wall distance and valid GPS can provide external references. IMU integration alone cannot guarantee lateral position. No before/after drift numbers were obtained, so closed-loop drift correction is not proven.

9. Test matrix and gates

Test	Status	Evidence/reason
Source audit	PASS	Handoff and source inspected
Main compile	PASS	Arduino CLI build
Camera compile	PASS	Stream compile defect fixed
Flash hash verification	PASS	esptool verified
Partition decode/readback	PASS	Valid table and stable reads
Main AP boot	FAIL	Brownout at Wi-Fi startup
Camera soak	NOT RUN	Main hub unavailable
Physical movement	NOT RUN	Must be physically confirmed
GPS outdoor run	NOT RUN	Main application not stable
Drift before/after	NOT RUN	No measured runs
Three outdoor runs	NOT RUN	Phase 1 gates not passed

Gates not met

- Heartbeat under 1000 ms during captures: not measured.

- 95% valid JPEGs and zero 4-byte files: not measured.
- Zero unexplained rover disconnects: not demonstrated.
- Physical movement: not demonstrated.
- Measured drift improvement: not measured.
- Complete valid ZIP each run: not demonstrated.
- Three consecutive outdoor runs: not performed.

10. Remaining risks and next actions

Priority	Action	Pass condition
P0	Stable regulated power/current to closed box; do not rely on USB-UART	40 MHz minimal test prints POWER_TEST_ALIVE continuously
P0	If known-good power still fails, test replacement ESP32/regulator/rail	Minimal test stays alive on known-good hardware
P1	Use explicit 40 MHz production builds	No checksum corruption and stable AP
P1	Run full firmware serial/AP/screen test	CAS-Box-04 and 192.168.14.1 remain available
P1	Camera registration/snapshot test	Fresh valid JPEGs through relay
P1	Wheels-off-ground safety/movement test	Physical rotation, deadman, latch and heartbeat pass
P2	Telemetry/image soak and ZIP reconciliation	Counts and references reconcile
P2	GPS outdoor run	Valid/invalid GPS states preserved
P2	Drift baseline/correction measurement	Before/after heading/lateral numbers
P3	Only after Phase 1, tune outdoor thresholds	Three consecutive outdoor runs pass

Do not: disable brownout, weaken deadman/heartbeat/latch/obstacle stop, treat HTTP 200 as movement proof, fabricate GPS, or change partitions as a response to confirmed brownout.

Appendix: files, commands and provenance

Path	Purpose
cas_box/cas_box.ino	Main ESP32 setup/loop

cas_box/Sensors.cpp/.h	LiDAR, IMU, GPS and ultrasonic
cas_box/RoverLink.cpp/.h	Main-to-rover link
cas_box/WebHMI.cpp/.h	AP, UI, camera relay and drive API
camera_s3/CameraStation/CameraStation.ino	Camera snapshot server
tools/build_r04.sh	Repeatable Arduino CLI build
tools/diagnose_main_esp32.py	Read-only diagnostic bundle
tools/power_test/power_test.ino	Minimal Wi-Fi isolation sketch
flash/*.bin	Downloadable merged images

Diagnostic command

```
python3 tools/diagnose_main_esp32.py --port /dev/ttyUSB0 --expected-bin build/box/cas_box.ino.bin --out diagnostic_main_esp32
```

Final evidence statement

Source-level camera and phone changes were implemented and compiled. The connected board was identified, the partition table decoded, flash readback repeated, and the minimal Wi-Fi test isolated the current failure to brownout at Wi-Fi startup. The system is not declared operational; Phase 1 and Phase 2 definition-of-done gates remain open.

Generated from repository commits and terminal/serial evidence available in this workspace. Where a hardware test was not run, this report says so explicitly.